

Lecture 4: Parallel Lines and Parallelograms

Definition of Parallel Lines

Two lines are **parallel** if they do not intersect (i.e., they have no common point).

Euclid's fifth postulate (Parallel Postulate)

Given a line ℓ and a point P not on ℓ , there is exactly one line through P that is parallel to ℓ .

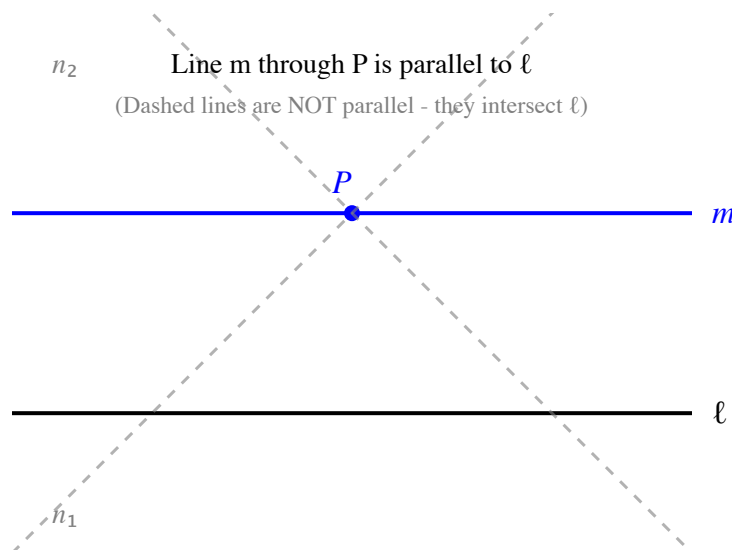


Figure 1: Euclid's fifth postulate

The two dashed lines in the picture are nonparallel to ℓ because they intersect ℓ . The solid line is the unique line through P that is parallel to ℓ .

Parallel lines can be constructed using skidding rulers.

Basic Properties of Parallel Lines

We use $a \parallel b$ to denote two lines a and b are parallel. 1. If $a \parallel b$ and $b \parallel c$, then $a \parallel c$. (Transitivity) 2. If $a \parallel b$ then $b \parallel a$. (Symmetry) 3. $a \parallel a$. (Reflexivity)

The transitivity property requires a proof, while the other two properties are obvious. If $a \parallel b$ and $b \parallel c$, but a is not parallel to c , then a and c intersect at some point P . By Euclid's fifth, there is exactly one line through P parallel to b . But both a and c are parallel to b and pass through P , which is a contradiction.

Equality of segments on parallel lines

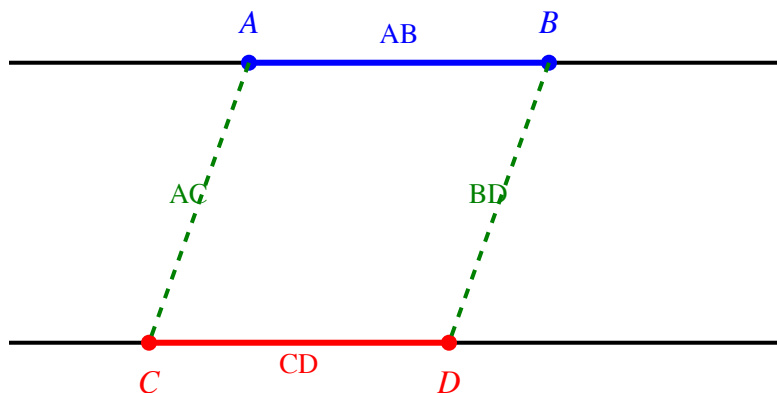
Definition: Two segments AB and CD are **equal** on two parallel lines if AC and BD (or AD and BC) are parallel.

Definition: A **parallelogram** is a quadrilateral $ABCD$ such that $AB \parallel CD$ and $AD \parallel BC$.

Key property: two opposite sides of a parallelogram are equal, i.e., $AB = CD$ and $AD = BC$.

Construction #1: Given a segment AB , a point C on a line ℓ parallel to AB , we can first construct the line AC , then construct the line through B parallel to AC to find point D such that $ABCD$ is a parallelogram and $AB = CD$.

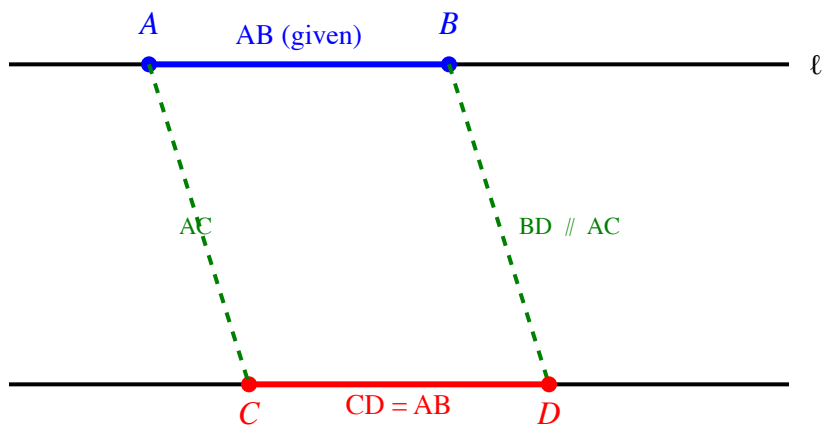
Equal Segments on Parallel Lines



$AB = CD$ because $AC \parallel BD$

Figure 2: Equal segments on parallel lines

Construction #1: Parallelogram ABDC



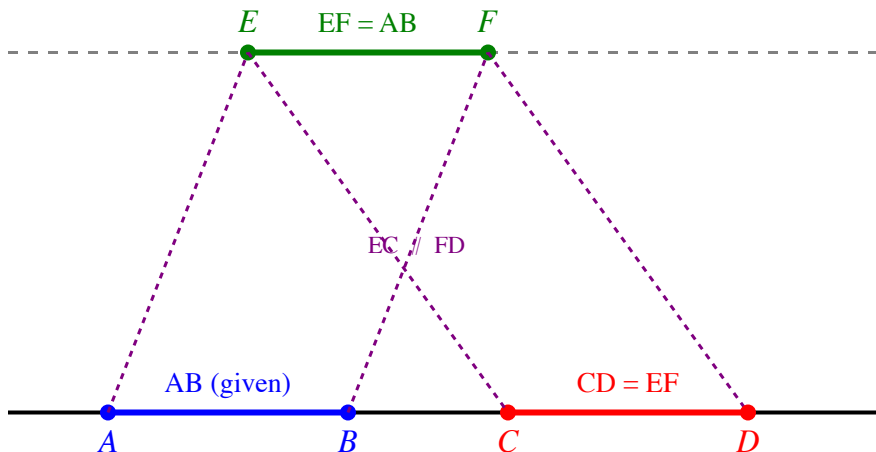
Given AB and point C on $\ell \parallel AB$, construct D so that $CD = AB$

Figure 3: Construction of parallelogram

Construction #2: Given a segment AB and a point C on the same line, construct a line segment CD such that $AD = BC$.

By construction #1, we can first construct a parallel line to AB and a line segment EF equal to AB . Then by construction #1 again, we can construct $CD = EF$.

Construction #2: Equal Segments on Same Line



Given AB and C on same line, construct D such that $CD = AB$
 (Uses Construction #1 twice: first for $EF = AB$, then for $CD = EF$)

Figure 4: Construction of equal segments

Construction #3: Given a segment AB , construct a line segment on the same line doubling the length. This is easy by putting $C = B$.

Construction #4: Halving a segment

Given a segment AB , construct its midpoint M such that $AM = MB$.

Steps: 1. Construct another line through A 2. Pick a point C on this line and construct AC . 3. By Construction #3, construct a point D such that $AC = CD$. 4. Construct the line BD . 5. Construct the line through C parallel to BD and let it intersect AB at M .

Key Lemma: Equal segments on a line projects to equal segments on another line via parallel lines.

We will prove this in next lecture.

Bimedian of a Triangle

Given a triangle ABC , the **bimedian** is the segment determined by the parallel line through the midpoint of one edge.

Theorem: The bimedian of the triangle ABC is half the length of the opposite edge, i.e., $MN = \frac{1}{2}BC$.

Construction #3: Double the Length

Given AB , set $C = B$, then construct D such that $BD = AB$

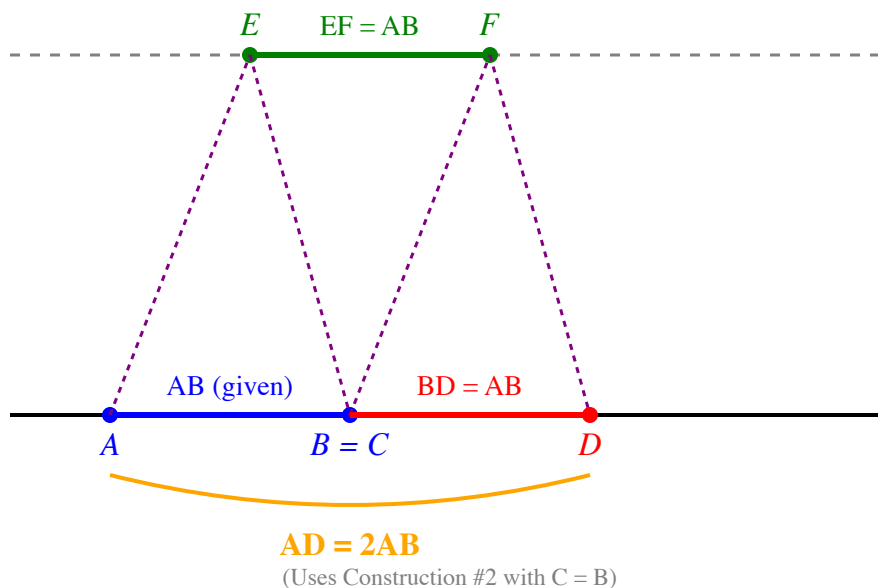
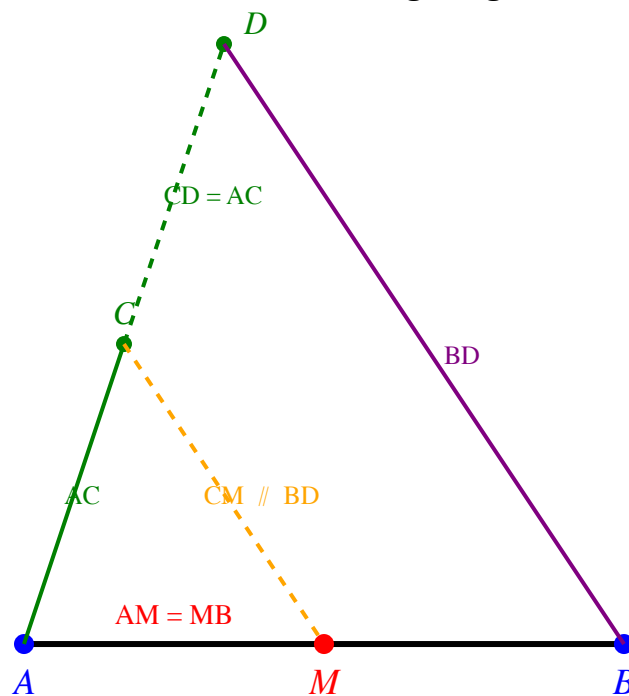


Figure 5: Construction of double segment

Steps of the proof: 1. Construct the midpoint of AB by Construction #4 and denote it by M . 2. Construct the line through M parallel to BC and let it intersect AC at N . 3. By the key lemma, $AM = MB$ implies $AN = NC$. Thus, N is the midpoint of BC . 4. Construct the bimedial NO through N parallel to AB and let it intersect AC at O . 5. By the key lemma again, $AM = MB$ implies $NO = OC$. Thus, O is the midpoint of AC . 6. Finally, since $MNOB$ is a parallelogram, we have $MN = OB$. But $OB = \frac{1}{2}BC$ since O is the midpoint of BC . Therefore, $MN = \frac{1}{2}BC$.

Construction #4: Halving a Segment

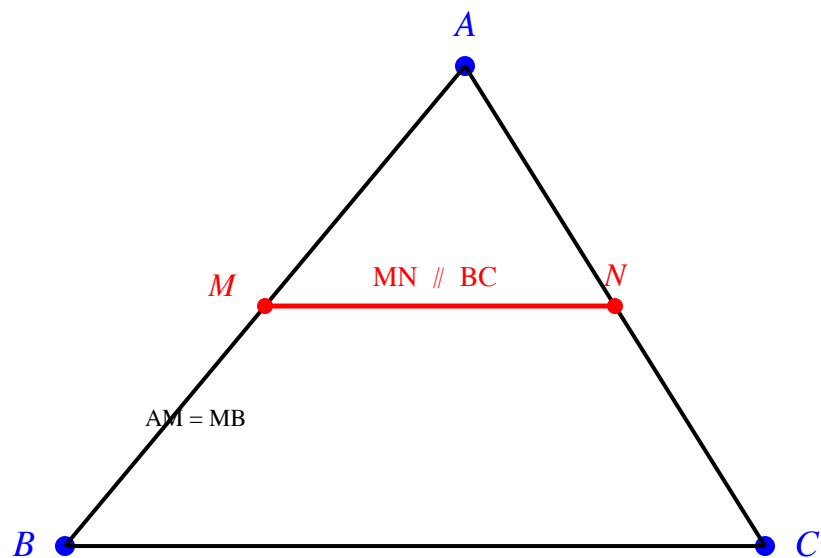


Given AB, construct midpoint M such that $AM = MB$

Line through C parallel to BD intersects AB at midpoint M

Figure 6: Construction of midpoint

Bimedian of a Triangle



M is the midpoint of AB ; MN is parallel to BC (horizontal)

Theorem: $MN = \frac{1}{2}BC$

Figure 7: Bimedian of a triangle

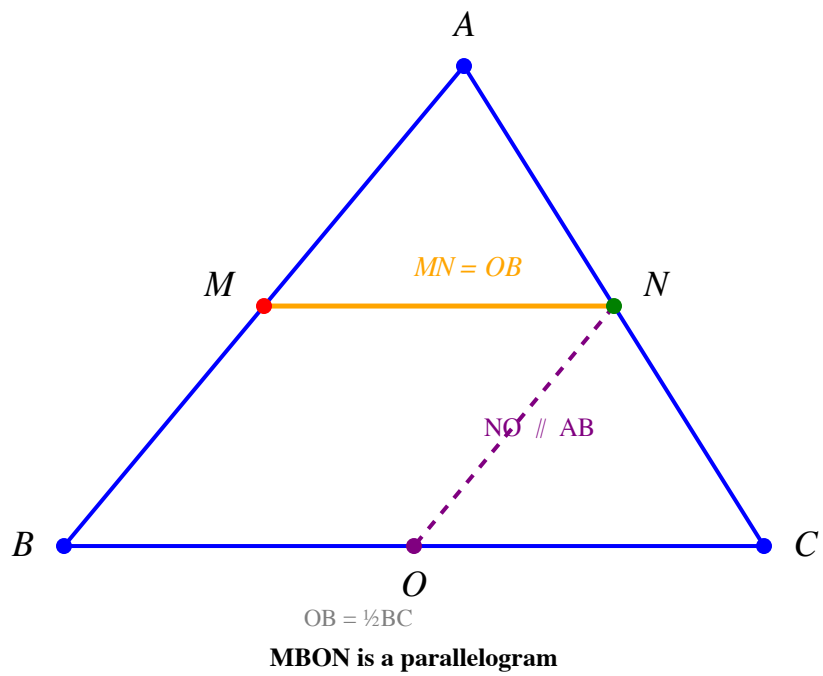


Figure 8: Bimedial theorem